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Housing Cost Burden and Economic Vulnerability in California

DSCI 510 | Spring 2026 | USC

Final Project **Presentation**

Project Introduction

This project builds a data pipeline analyzing housing cost burden across California's 58 counties from 2012 to 2023, combining Census ACS, HUD Fair Market Rents, and BLS labor data to group counties into four policy-relevant tiers.

Data Sources

Source	What It Provides	Processing
Census ACS	Median gross rent and renter burden count by county	API pull, missing values replaced with NaN, FIPS extended to 9 digits to match HUD
HUD Fair Market Rents	2-bedroom fair market rent by county	Loaded from a single historical CSV, filtered to California, reshaped from wide to long format by year
BLS LAUS	County-level unemployment rates	Loaded from CSV, filtered to county rows and relevant years, county name cleaned for merging

All three merged at the county-year level using FIPS codes and county name.

Final dataset: 693 county-year observations across 58 California counties and 12 years (2012–2023), after inner-joining all three sources and dropping rows with missing values.

Clustering Features

Feature	Description
Median Gross Rent	Typical monthly rent paid by renters in a county
HUD 2BR Fair Market Rent	HUD's benchmark affordable rent for a 2-bedroom unit
Unemployment Rate	Share of labor force without a job

Cluster Summary

4 Policy Tiers

High Rent Burden

14 counties. Avg. rent $2,204, unemployment 4%. Bay Area and Central Coast counties where cost far outpaces wages. Even employed residents are spending a disproportionate share of income on rent.

Moderate Rent Burden

15 counties. Avg. rent $1,605, unemployment 5%. Large metros with rising rents and stable labor markets. These counties are trending toward the high rent burden tier

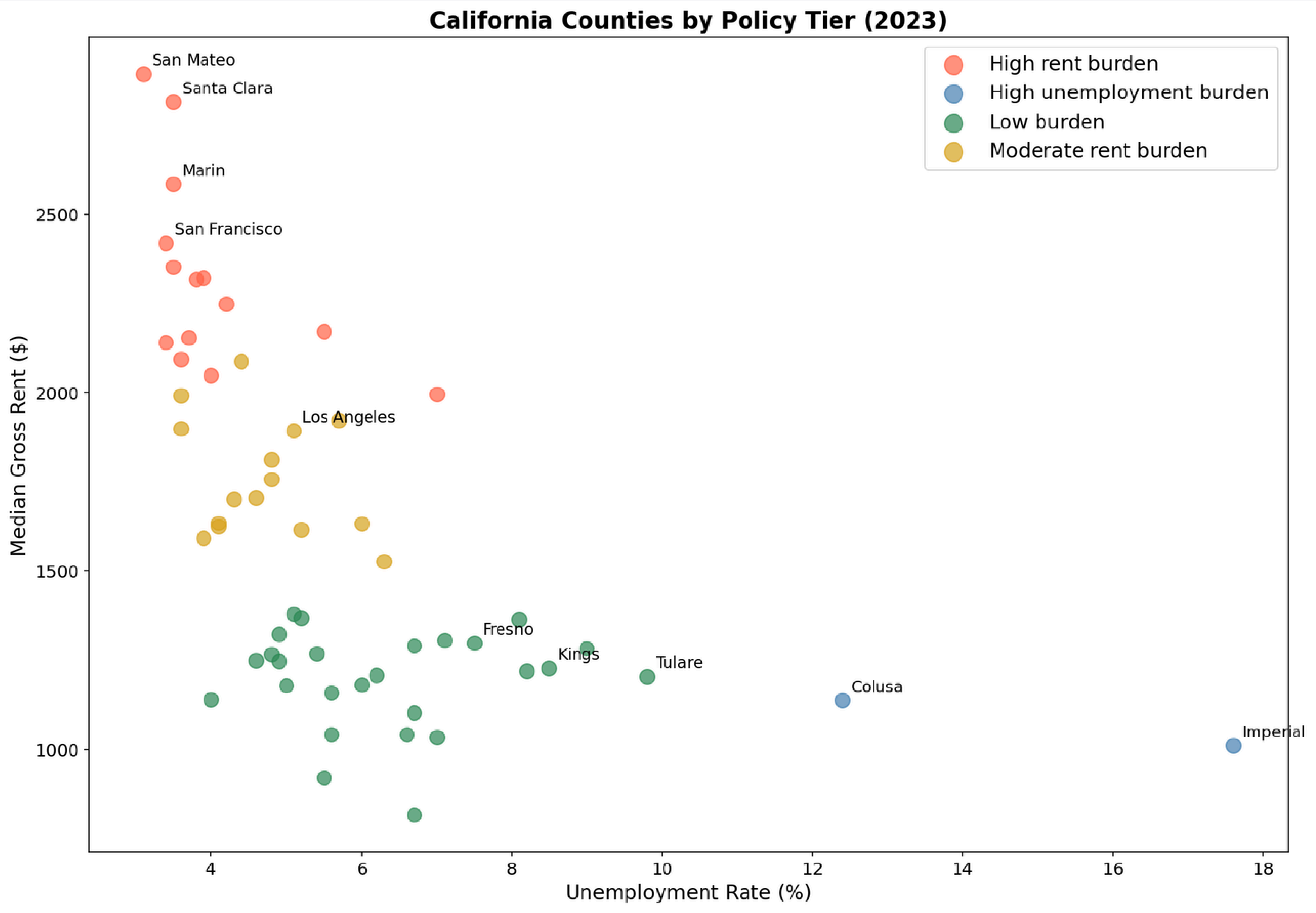
Low Burden

26 counties. Avg. rent $1,061, unemployment 7%. Inland and rural counties. Lower rents, but wages are also lower. The label is relative to the rest of the state, not a statement that housing is affordable.

High Unemployment Burden

2 counties. Colusa and Imperial. Rents are low but unemployment averages 13%, meaning even relatively low housing costs are out of reach for many residents. A structurally different problem from the other three tiers.

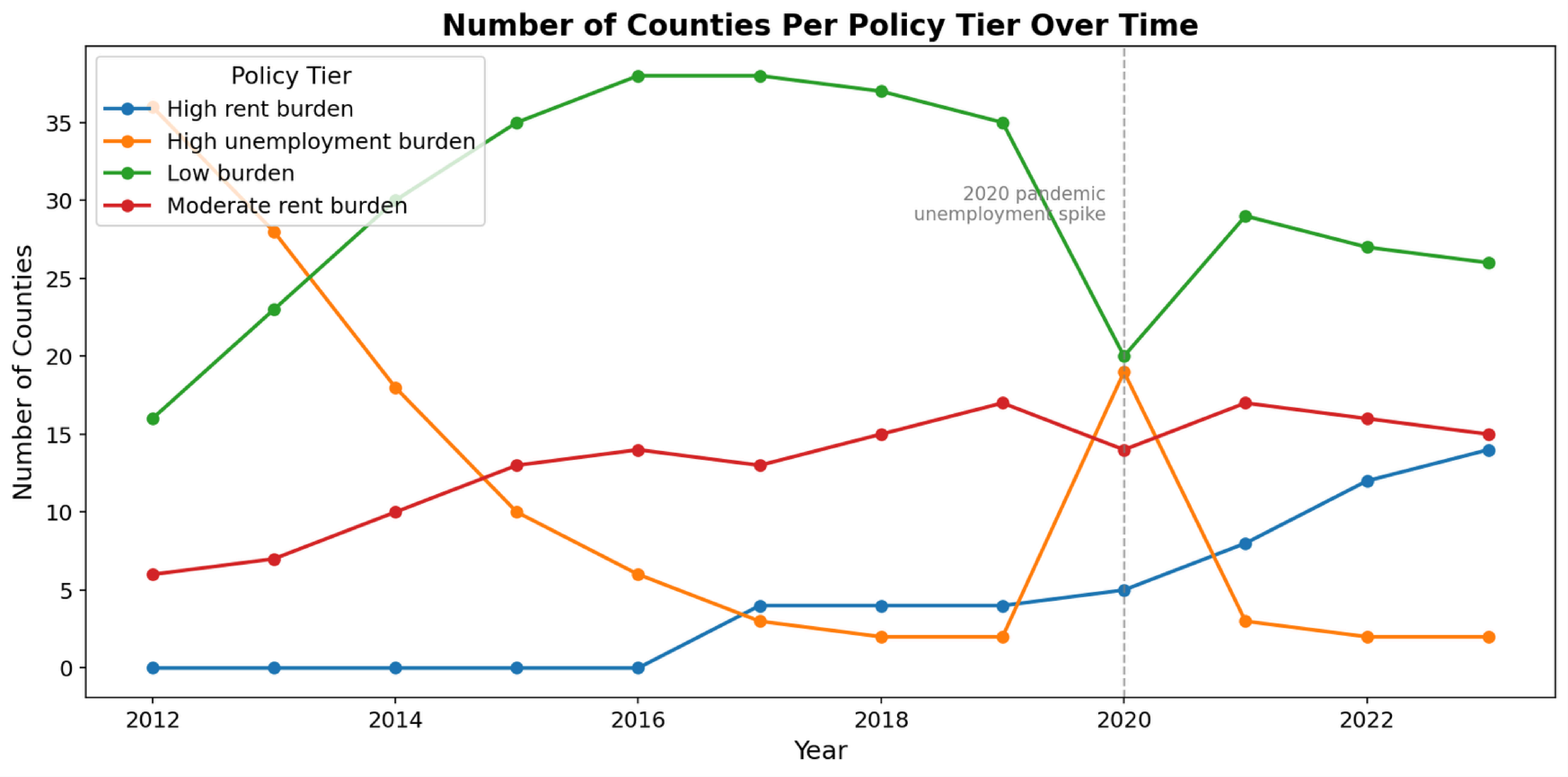
2023 County Scatter Plot



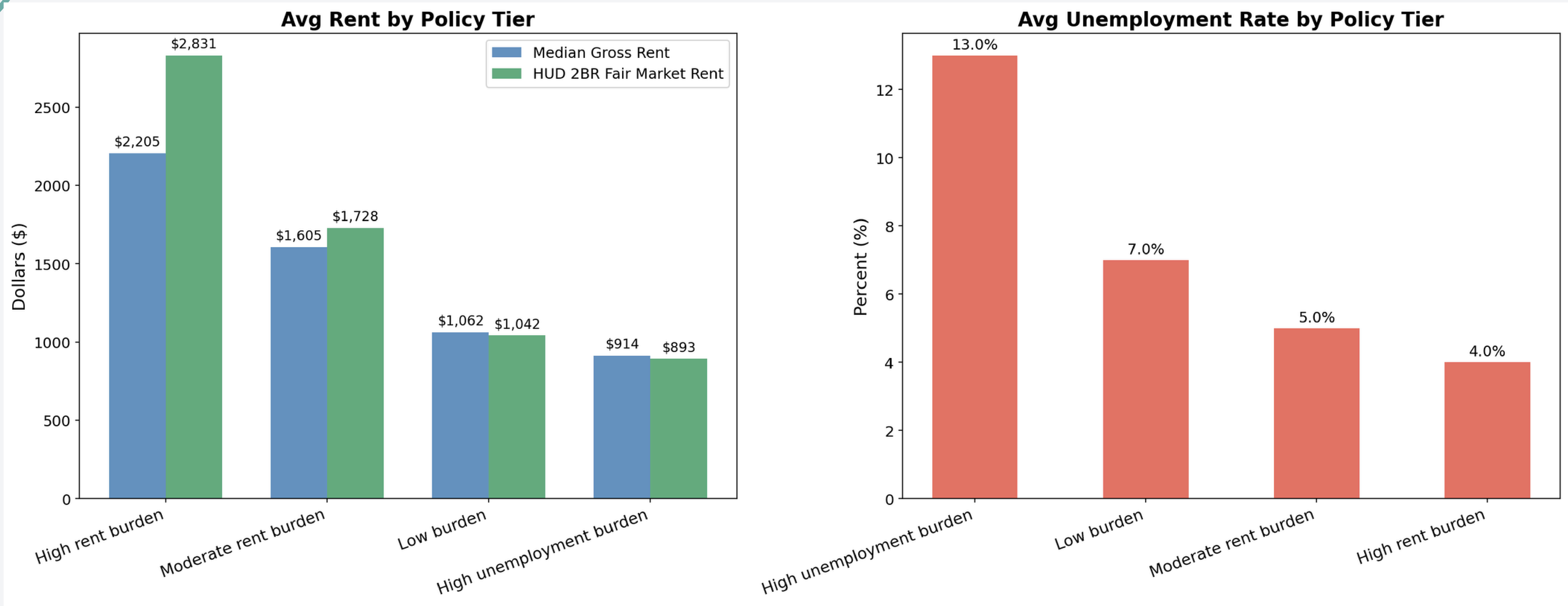
Each point is one county in 2023, colored by policy tier. Bay Area counties cluster in the upper left with high rent, low unemployment. Los Angeles sits in the moderate tier at around $1,900 rent and 6% unemployment. Colusa and Imperial are isolated far right with high unemployment and low rents.

Tier Over Time

In 2012, 36 counties were classified as high unemployment burden following the Great Recession. Counties shifted steadily toward low burden and moderate rent burden as rents rose and labor markets recovered. The high rent burden tier grew from near zero in 2012 to 14 counties by 2023. The 2020 pandemic caused a sharp spike before a rapid recovery through 2022.



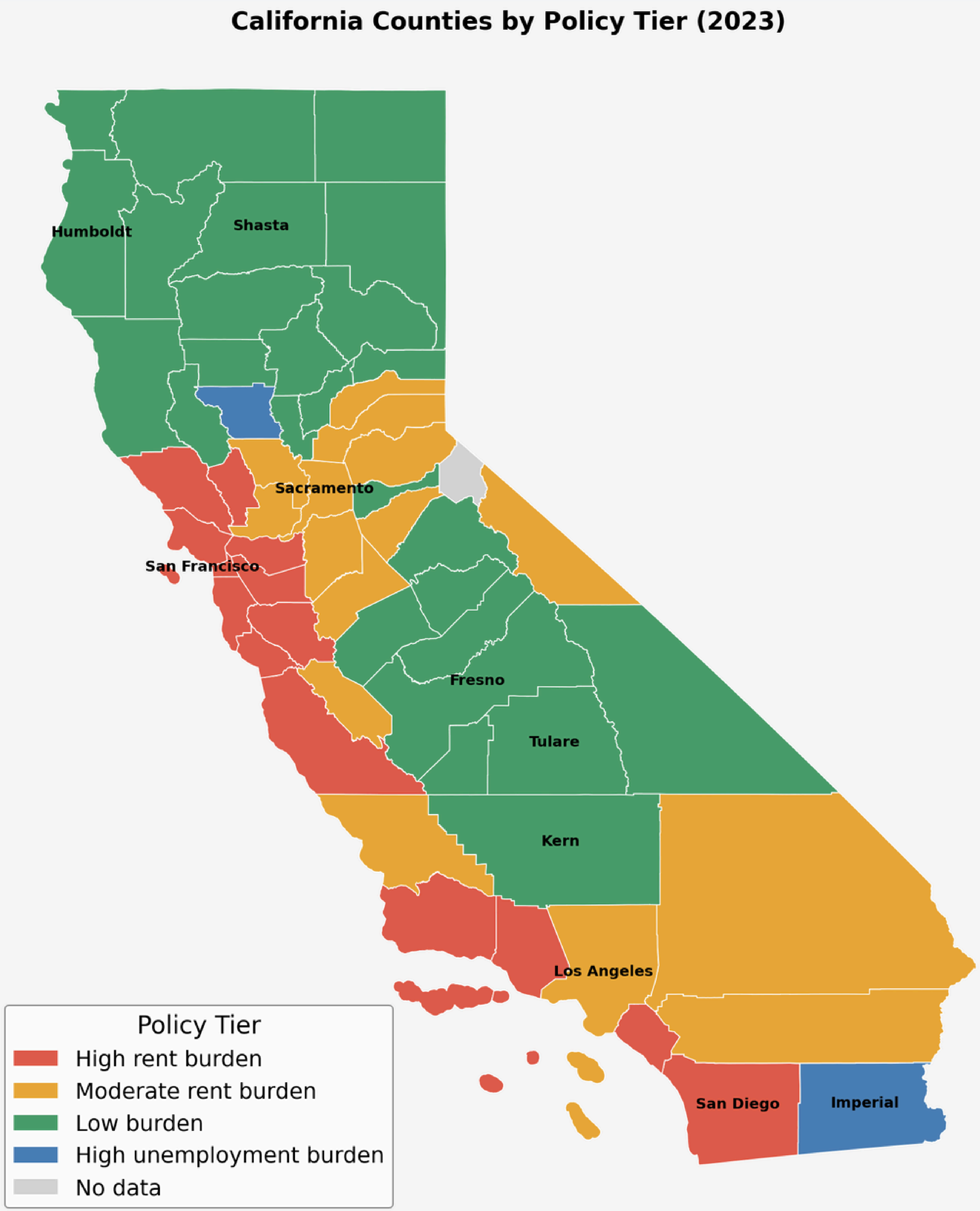
Cluster Summary Chart



The left panel compares average rent across tiers, stepping down from $2,204 in the high rent burden tier to $913 in the high unemployment burden tier. The right panel shows the inverse pattern on unemployment, with high unemployment burden counties averaging 13% compared to just 4% for high rent burden counties.

California County Map

High rent burden runs along the Bay Area and Central Coast, including San Diego at $2,154 median gross rent and 3.7% unemployment rate in 2023. Moderate rent burden covers the suburban inland ring including Los Angeles, Sacramento, and the Inland Empire. Low burden dominates the rural North and Central Valley. Colusa and Imperial are the only two high unemployment burden counties remaining in 2023.



Project Challenges

- **Data Integration:** Three datasets with different county identifier formats required standardization before merging.
- **Missing Values:** The Census API flags missing data with a numeric code rather than a blank, which had to be caught before analysis.
- **Wide to Long Format:** HUD stored one column per year across hundreds of columns. Reshaping to one row per county per year was required before merging.
- **Clustering:** K=3 produced a lopsided split with one tier containing only 2 counties. K=4 gave a more balanced and interpretable result.
- **Future Directions:** A key limitation is the absence of housing supply data. Adding county-level housing stock or vacancy rates would allow for a more complete analysis of affordability beyond rent prices and unemployment alone.

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Thank You!

Housing Cost Burden Analysis